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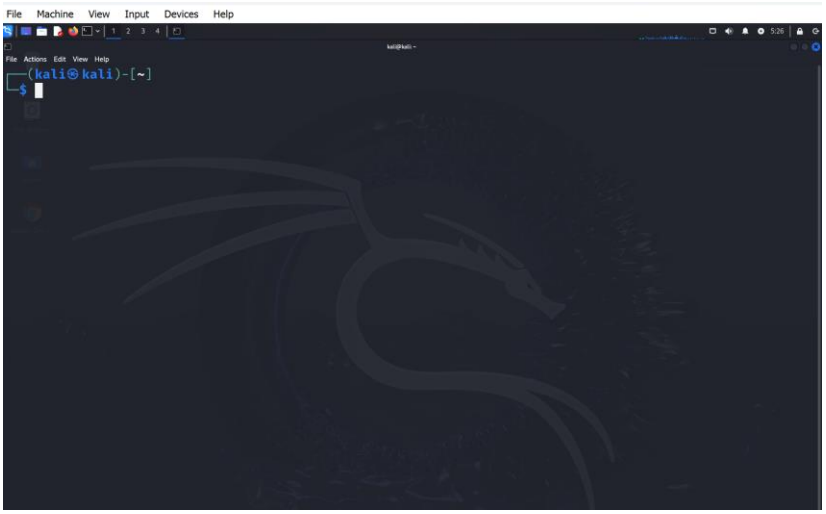
CYBI 3345

Lab 05- Linux Security Basics

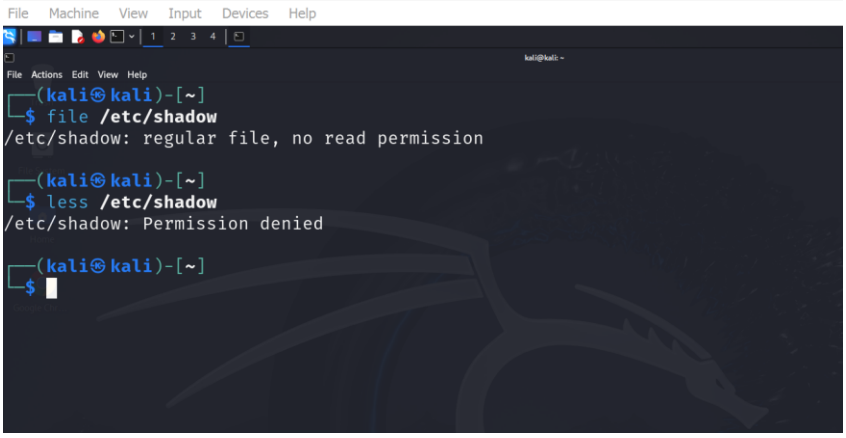
Introduction: In this lab, I will use basics commands to strengthen my understanding of security basics in the Linux OS.

Objective: To manipulate files and directories permissions in Linux

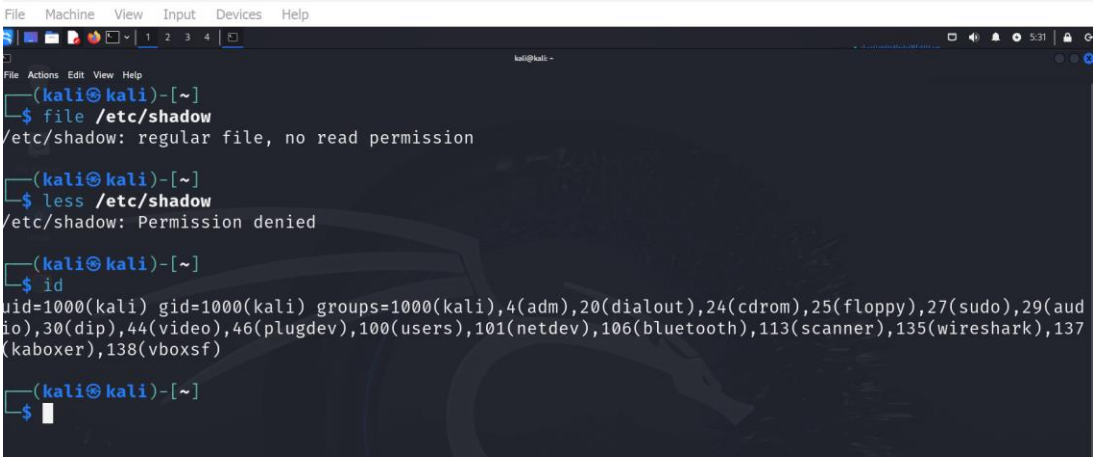
Step 1: For this step, I opened a terminal by using the ctrl+alt+t shortcut, it just opened a terminal.



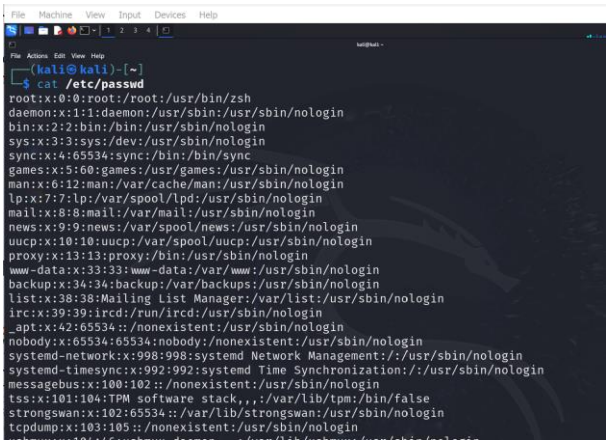
Step 2: In this step,I used the command >>/etc/shadow<< it says that permission was denied.

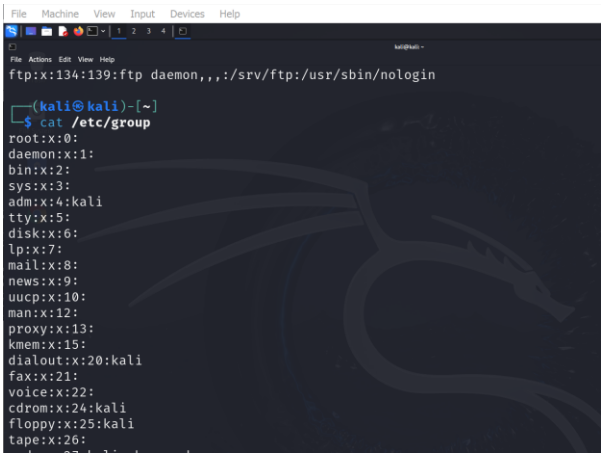


Step 3: The id command showed me the following process ID.



Step 4: The commands helped me in see the user accounts. After using the given commands, the user is kali. I used the >>cat /etc/group<< command.





Step 5: After using the ls –l command, it helped me visualize the permissions for a file r directory. The command >>ls –l<< was used.

```
┌──(kali㉿kali)-[~]
└─$ foo.txt
oo.txt: command not found

┌──(kali㉿kali)-[~]
└─$ > foo.txt
s -l foo.txt
C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
rw-r--r-- 1 kali kali 14 May  2 05:44 foo.txt

┌──(kali㉿kali)-[~]
└─$
```

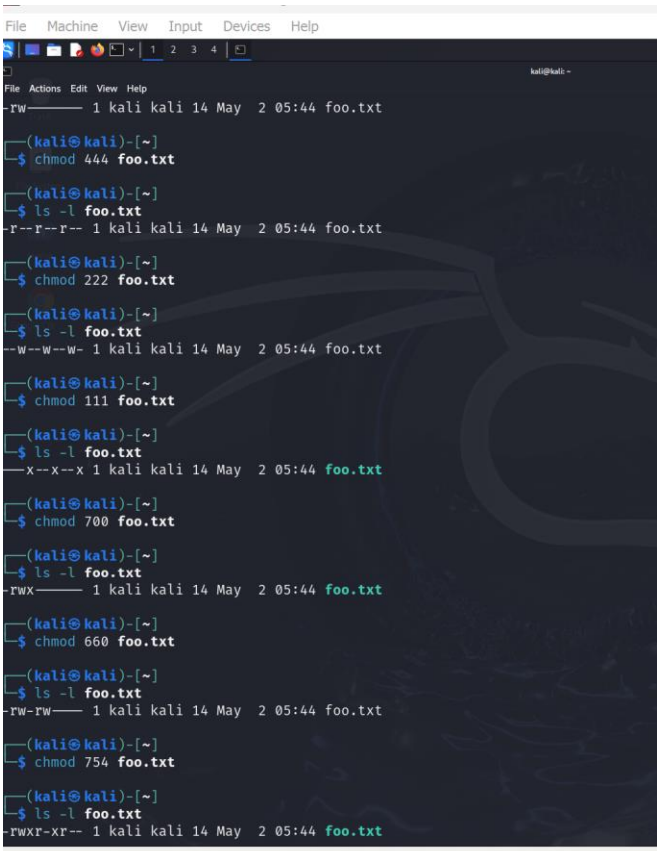
Step 6: Now, the permissions have changed after using the chmod 600 command.

```
┌──(kali㉿kali)-[~]
└─$ chmod 600 foo.txt

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
rw------- 1 kali kali 14 May  2 05:44 foo.txt

┌──(kali㉿kali)-[~]
└─$
```

Step 7: Here are the commands using the octal and symbolic representation. I checked that the change had happened by using the ls –l command to view the permissions.



Step 8: In this step, I used the rm –r command to delete any similar stuff to make sure I started fresh. Then, I used the umask to view its content. After that, I used the ls –l to view the permissions.

```
┌──(kali㉿kali)-[~]
└─$ rm -f foo.txt

┌──(kali㉿kali)-[~]
└─$ umask
022

┌──(kali㉿kali)-[~]
└─$ > foo.txt
^C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
-rw-r--r-- 1 kali kali 0 May  2 05:59 foo.txt

┌──(kali㉿kali)-[~]
└─$
```

Step 9: I changed the permission and now the file has been turned off.

```
┌──(kali㉿kali)-[~]
└─$ rm foo.txt

┌──(kali㉿kali)-[~]
└─$ umask 0000

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
ls: cannot access 'foo.txt': No such file or directory

┌──(kali㉿kali)-[~]
└─$ > foo.txt
^C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
-rw-rw-rw- 1 kali kali 0 May  2 06:03 foo.txt

┌──(kali㉿kali)-[~]
└─$
```

Step 10: I used the rm and unmask commands and then the 0022, 0777 and 0000.

```
┌──(kali㉿kali)-[~]
└─$ rm foo.txt

┌──(kali㉿kali)-[~]
└─$ umask 0022

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
s: cannot access 'foo.txt': No such file or directory

┌──(kali㉿kali)-[~]
└─$ > foo.txt
C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
rw-r--r-- 1 kali kali 0 May  2 06:07 foo.txt
```

```
┌──(kali㉿kali)-[~]
└─$ rm foo.txt

┌──(kali㉿kali)-[~]
└─$ umask 0777

┌──(kali㉿kali)-[~]
└─$ > foo.txt
^C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
---------- 1 kali kali 0 May  2 06:08 foo.txt

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ rm foo.txt
m: remove write-protected regular empty file 'foo.txt'? y

┌──(kali㉿kali)-[~]
└─$ umask 0000

┌──(kali㉿kali)-[~]
└─$ > foo.txt

^C

┌──(kali㉿kali)-[~]
└─$ ls -l foo.txt
rw-rw-rw- 1 kali kali 1 May  2 06:09 foo.txt

┌──(kali㉿kali)-[~]
```

Questions:

1) chmod: It is used to change the permissions (read, write, execute) of a file or directory. It can be used to set permissions for the owner, group, and others.

umask: It is a value that is subtracted from the default permissions assigned to newly created files and directories. It sets permissions that are not granted by default. For example, if the umask is set to 022, new files will have permissions of 644 (666 - 022) and new directories will have permissions of 755 (777 - 022).

2) The most common and convenient umask value to protect files is 077. This setting ensures that new files are only accessible to the owner and not to any other user or group. For directories, a umask value of 022 or 027 is often used to allow group access but prevent others from accessing or modifying the directory contents.

3) Yes, you can set permissions to a specific value without using the command line by using a file manager or GUI tool that allows you to change permissions. For example, in Linux, you can use the file manager's properties dialog to set the permissions graphically.

4) Username: The name of the user.

Password: An 'x' character indicating that the encrypted password is stored in the /etc/shadow file.

User ID (UID): The unique numerical ID assigned to the user.

Group ID (GID): The numerical ID of the primary group of the user.

User ID Info: Additional information about the user (e.g., full name).

Home directory: The path to the user's home directory.

Shell: The path to the user's default shell program.

Conclusion: In this lab, I learned security basics in the Linux OS. I used the umask, learned the differences between octal and symbolic representation. I also used to say commands on a file by visualizing the permission status. Overall, I have a better understanding of security basics.

References:

"Protect Your Personal Linux/Unix Computer." U, safecomputing.umich.edu/protect-yourself/secure-your-devices/personal-computer/linuxunix. Accessed 2 May 2024.

Zivanov, Sara. "Linux Security Stats, Tools, and Best Practices: Phoenixnap." Knowledge Base by phoenixNAP, 12 Feb. 2024, phoenixnap.com/kb/linux-security.